

NOT Θ

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Let $\mathbf{D}$ be the 2-category whose underlying category is $\Delta_{\perp \neq \top}$ and whose 2-cells are pointwise order. We regard $\mathbf{D}$ as a locally full sub-2-category of $\mathbf{Cat}$. Many of our examples of $\mathcal{P}$ are also naturally equipped with a locally ordered 2-category structure. Moreover, the appropriate choice of m^* for $m \in \mathcal{M}$ is often a left adjoint in $\mathcal{P}$ with identity counit (called a *lali*).

There is a kind of wreath product $\mathcal{Q}$ of $\mathbf{D}$ with $\mathcal{P}$. It is another locally ordered 2-category. The objects are functors $A : a \rightarrow \mathcal{P}$ (not just families) with $a \in \mathbf{D}$. A morphism $(\xi, u) : (a, A) \rightarrow (b, B)$ is a diagram

$$\begin{array}{ccc} a & \xrightarrow{\xi} & b \\ & \searrow A \quad \xRightarrow{u} \quad \swarrow B & \\ & \mathcal{P} & \end{array} \quad (0.1)$$

in $\mathbf{Cat}$ with $\xi : a \rightarrow b$ in $\mathbf{D}$. Define $(\xi, u) \leq (\zeta, v) : (a, A) \rightarrow (b, B)$ when $\xi \leq \zeta$ and, for each $i \in a$, we have $B(\xi i \leq \zeta i) \circ u_i \leq v_i$.

In other words, $\mathcal{Q}$ is obtained by the Grothendieck fibration construction applied to the 2-functor

$$\Delta_{\perp \neq \top}^{\text{op}} \longrightarrow \mathbf{Cat}^{\text{op}} \xrightarrow{[-, \mathcal{P}]} \mathbf{Cat} .$$

Define $\mathcal{M}$ for $\mathcal{Q}$ to consist of the morphisms $(\partial, m) : (a, A) \rightarrow (b, B)$ where ∂ is injective and $m_i \in \mathcal{M}$ for all $i \in a$.

Proposition 0.1. *If σ is a lali for ∂ in $\mathbf{D}$ and m_i^* is a lali in $\mathcal{P}$ for $m_i \in \mathcal{M}$, $i \in a$, then (σ, n) is a lali for (∂, m) , where $n_k = m_{\sigma k}^* \circ B(k \leq \partial \sigma k)$ for all $k \in b$.*

In this way, we get a functorial choice of $(\partial, m)^*$ for all $(\partial, m) \in \mathcal{M}$.

Let us assume that each object of $\mathcal{P}$ has only finitely many $(\mathcal{M})$ -subobjects and finite intersections of subobjects exist (as fibre products in $\mathcal{P}$).

Proposition 0.2. *Each object of $\mathcal{Q}$ has only finitely many subobjects and finite intersections of subobjects exist.*

Proof. The first part is clear since it is true of $\Delta_{\perp \neq \top}$ and assumed of $\mathcal{P}$. The pullback of two subobjects $(\partial, m) : (a, A) \rightarrow (c, C)$ and $(\partial', m') : (b, B) \rightarrow (c, C)$ is obtained by forming the pullback

$$\begin{array}{ccc} p & \xrightarrow{\pi'} & b \\ \pi \downarrow & & \downarrow \partial' \\ a & \xrightarrow{\partial} & c \end{array}$$

in $\Delta_{\perp \neq \top}$, and, for each $k \in p$, forming the pullback

$$\begin{array}{ccc} P_k & \xrightarrow{\ell'_k} & B_{\pi'k} \\ \ell_k \downarrow & & \downarrow m'_{\pi'k} \\ A_{\pi k} & \xrightarrow{m_{\pi k}} & C_{\partial \pi k} \end{array}$$

in $\mathcal{P}$. This gives the pullback

$$\begin{array}{ccc} (p, P) & \xrightarrow{(\pi', \ell')} & (b, B) \\ (\pi, \ell) \downarrow & & \downarrow (\partial', m') \\ (a, A) & \xrightarrow{(\partial, m)} & (c, C) \end{array}$$

in $\mathcal{Q}$. □

Defining $\mathcal{S}$ for $\mathcal{Q}$ to consist of those morphisms factoring through no proper sub-object, we conclude from the last proposition that we have the factorization property $(\xi, u) = (\partial, m) \circ (\sigma, s)$ with $(\partial, m) \in \mathcal{M}$ and $(\sigma, s) \in \mathcal{S}$.

Before continuing, we require a more explicit description of the $\mathcal{S}$ for $\mathcal{Q}$.

Proposition 0.3. *A morphism $(\xi, u) : (a, A) \longrightarrow (b, B)$ in $\mathcal{Q}$ is in $\mathcal{S}$ if and only if ξ is surjective and, for each $j \in b$, there exists $i \in a$ with $\sigma i = j$ and $u_i \in \mathcal{S}$.*

Proof. To prove “if”, assume $(\xi, u) = (\partial, m) \circ (\zeta, v)$ with $(\partial, m) \in \mathcal{M}$. Since $\xi = \partial \circ \zeta$ is surjective, ∂ is an identity and $\xi = \zeta$. So we have $u_i = m_{\xi i} \circ v_i$ for all $i \in a$. For any given $j \in b$, choose i in the fibre of ξ over j with $u_i \in \mathcal{S}$. Then $m_j = m_{\xi i}$ is invertible. This proves (∂, m) invertible. So $(\xi, u) \in \mathcal{S}$.

For “only if”, first note that, writing $\xi = \mu \circ \sigma$ with μ injective and σ surjective, we have $(\xi, u) = (\mu, 1_{B_\mu}) \circ (\sigma, u)$ with $(\mu, 1_{B_\mu}) \in \mathcal{M}$. So μ is an identity and ξ is surjective. Now notice that, by naturality of u , if $i_1 \leq i_2$ in the fibre of ξ over $j \in b$, then $u_{i_1} = u_{i_2} \circ A(i_1 \leq i_2)$. So $u_{i_1} \in \mathcal{S}$ implies $u_{i_2} \in \mathcal{S}$. Let ∂j denote the largest element of the fibre of ξ over j . So we are required to prove that each $u_{\partial j}$ is in $\mathcal{S}$. (In fact, $\xi \dashv \partial$ in $\mathbf{Cat}$ although $\partial : b \rightarrow a$ may not be in $\Delta_{\perp \neq \top}$). We have $(\xi, w) : A \longrightarrow (b, A_\partial)$ defined by $w_i = A(i \leq \partial \xi i) : A_i \longrightarrow A_{\partial \xi i}$. However, $(\xi, u) = (1_b, u_\partial) \circ (\xi, w)$ and $(\xi, u) \in \mathcal{S}$ imply $(1_b, u_\partial) \in \mathcal{S}$. It remains to show that any morphism in $\mathcal{S}$ of the form $(1_b, y) : (b, C) \longrightarrow (b, B)$ has each y_j in $\mathcal{S}$. Factor $y_j = m_j \circ s_j$ with $m_j \in \mathcal{M}$ and $s_j \in \mathcal{S}$. Then $(1_b, y) = (1_b, m) \circ (1_b, s)$. So $(1_b, m)$ is invertible. Hence each m_j is invertible yielding $y_j \in \mathcal{S}$. □

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